

Data Cleaning & Reporting Automation

Project Overview

This project focuses on automating data cleaning and reporting processes using Python. The objective is to improve data quality, reduce manual effort, and generate meaningful reports automatically.

Dataset

The dataset contains information about vehicle sales including car names, manufacturing year, selling price, present price, fuel type, seller type, transmission type, and ownership details.

Tools Used

Python, Pandas, OpenPyXL, Matplotlib

Data Cleaning Tasks Performed

1. Removed duplicate records.
2. Handled missing values.
3. Verified dataset consistency.
4. Generated cleaned dataset automatically.

Reporting Automation

Automated reports were generated to provide total number of vehicles, average selling price, maximum selling price, and minimum selling price.

Visual Summary

A bar chart was generated to visualize revenue distribution across different fuel types.

Outcome

The project successfully automated the process of data cleaning and report generation, reducing manual effort and improving reporting efficiency.

Conclusion

Automation techniques can significantly improve data quality, save time, and support data-driven decision-making through consistent and reliable reporting.